INCH-POUND

A-A-59839
23 October 2009

COMMERICIAL ITEM DESCRIPTION

DECADE RESISTANCE STANDARD

The General Services Administration has authorized the use of the Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a wide range, four-terminal, Decade Resistance Standard with trimming capability. This device provides highly accurate resistance values from 10 mΩ to 1.2 MΩ with a minimum resolution of 20 μΩ. This CID lists the minimum performance, design, functional environmental, reliability requirements and product conformance provisions for this Decade Resistance Standard.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and bid sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 4 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.2.1 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated as a calibration standard at a temperature of 23 °C ± 1 °C. Furthermore this equipment shall conform to the specified temperature adjusted performance and accuracy while being operated in a minimum temperature range of 18 °C to 28 °C.

2.2.2 Temperature Non-Operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in a temperature range of ≤ -10 °C to ≥ 60 °C.

2.2.3 Humidity Range Operating. The equipment shall conform to its specified performance and accuracy while being operated in a non-condensing humidity range of ≤ 30% to ≥ 70%.

2.3 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 2,000 hours of operation with a statistical certainty of 95%.

2.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to WR-ALC/742 CBSG/GBEC, 460 Richard Ray Blvd. STE 200, Robins AFB, GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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2.5 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions, and operating service condition as specified in MIL-PRF-28800.

2.5.1 Range. This equipment shall have a minimum range of 10 mΩ to 1.2 MΩ.

2.5.2 Resolution. This equipment shall have a resolution of 20 µΩ or better.

2.5.3 Number of Decades: This equipment shall have a minimum of eight decade switch.

2.5.4 Accuracy.

2.5.4.1 Initial Accuracy. This equipment shall have an initial untrimmed accuracy of ± (20 ppm ± 1 mΩ) or better.

2.5.4.2 Trimmed Accuracy. This equipment shall have a trimmed accuracy of ± 2 ppm or better.

2.5.4.3 Long Term Accuracy. The Long Term Accuracy if this equipment, after adjustment, shall be ± (20 ppm + 0.5 mΩ per year) or better.

2.5.5 Temperature Coefficient. This equipment shall have a Temperature Coefficient of ± 20 ppm/°C or better.

2.5.6 Resistance Power Coefficient. This equipment shall have a Resistance Power Coefficient of ± 1.0 ppm/mW/Step, or better.

2.5.7 Maximum Power Rating. The Maximum Power Rating of this equipment shall be greater than or equal to ± 1.0 Watt/step up to 5 W total, or 2.0 Amps current.

2.5.8 Breakdown Voltage. The breakdown voltage of this equipment shall be ≥ 1500 V Peak to the case.

2.5.9 Rack Mount. The equipment shall be capable of being rack-mounted in a standard EIA/ECA-310E equipment rack with an optional rack-mount conversion kit.

2.5.10 Dimensions. The width and depth of the equipment shall be compatible with mounting in an EIA/ECA-310E standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by EIA/ECA-310E to be 1.75 inches or 44.45 mm.

2.5.11 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a one person lift of 37 pounds, (16.8 kilograms).

2.6 Manuals. The equipment shall be delivered with operator, maintenance, calibration, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

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3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) subcontractors comply with the requirement.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave., Bldg 4, Section D, Philadelphia, PA 19111-5094 or at http://assist.daps.dla.mil .

6.1.2 The Federal Acquisition Regulation (FAR) may be obtained at http://www.gpoaccess.gov/cfr/index.html or available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 Electronics Industry Alliance (EIA). Hardcopy of EIA standards and technical publications are available from Global Engineering Documents, the exclusive distributor of EIA standards or at http://global.ihs.com . Telephone number: 1-800-854-7179.

6.2 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Product conformance provisions.
c. Packaging requirements.

6.3 Key words.

Calibration
Non-condensing humidity range
Trimming capability

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MILITARY INTERESTS:

Custodian:
Air Force – 84

Preparing activity:
Air Force – 84

Review activity:
Air Force – 99

Agent:
Air Force – 99

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